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Analysis of Annotated Text Corpora for the Construction of LPP Argumentation Graphs

Master 1 Intelligence Artificielle Distribuée

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Chapter 1

Introduction

1.1 Context

The formalization of argumentative reasoning is a central challenge in artificial intelligence. Humans reason by constructing arguments, confronting them with counter-arguments, and then resolving conflicts through priorities. Reproducing this type of reasoning is a significant challenge for symbolic artificial intelligence, especially when knowledge is incomplete, contradictory, or context-dependent.

This Research Study and Work is part of the ANR GRAIL project, *Generative aRgumentative Artificial InteLLigence* (ANR-25-CE23-5597), led by LIPADE, LLF, and IRIT. This project aims to develop neuro-symbolic AI agents capable of learning, revising, and communicating knowledge through structured arguments. The goal is to go beyond purely statistical models to obtain systems capable of justifying their reasoning in an explicit and verifiable manner.

The formalism studied is LPP/GORGIAS, for *Logic Programming with Priorities*, developed by Kakas, Moraitis, and Spanoudakis [1]. It organizes reasoning into three levels: **rule**, which derive conclusions from factual conditions; **prefer**, which establish priority between competing rules; and **meta_prefer**, which establish priority between competing preferences. This formalism has notably been applied to medical assistance, network security, and cognitive personal assistants.

Our work starts from a corpus of 17 datasets provided by the supervisors of this research project. These resources cover biomedical, legal, and deliberative texts, argumentative essays, and online discussions. They contain both raw texts and highly varied annotations depending on the dataset: some schemes are close to the labels **Claim/Premise**, **Support**, or **Attack**, while others use categories specific to their domain. They constitute a useful panorama of argument mining, but are not annotated in LPP format. Exploiting them for LPP/GORGIAS therefore requires identifying, in natural text, facts, conclusions, rules, conflicts, and priorities.

1.2 Research Question

A natural text does not directly yield an LPP structure. Building a representation compatible with GORGIAS requires identifying the Context, the Option, the **rule**, and then, when the text justifies it, the **prefer** and **meta_prefer**. This operation does not correspond to a simple conversion from existing schemes: a **Claim** is not always an Option, a **Support** relation is not always a **rule**, and the available annotations do not necessarily specify the Condition, the Effect, or the priorities. We therefore had to determine whether existing annotations could serve as a basis for an LPP/GORGIAS representation, or whether it was necessary to work directly on raw text.

The central question of this TER is therefore the following: how can we guide the transition from a raw argumentative text to an LPP representation compatible with GORGIAS? Since existing annotations are not sufficient for our problem, we worked directly on raw text, while using the analysis of datasets to define annotation criteria. This research question is accompanied by a second one: to what extent can this transition be assisted by an automatic tool? The main difficulty concerns levels 1 and 2: a **prefer** requires two conflicting **rule** and an explicit priority, while a **meta_prefer** assumes that a preference is itself overridden in a more specific context. These structures are rare and difficult to infer without human validation.

1.3 Objectives

The first objective is to analyze the 17 provided datasets in order to evaluate their relevance for LPP and to identify usable information for constructing Context, Option, **rule**, **prefer**, and **meta_prefer**. The second objective is to define annotation principles enabling the transition from the classical labels of argument mining to LPP levels, while making explicit the limits of the correspondences between **Claim**, **Premise**, **Support**, **Attack**, and the structures expected by GORGIAS.

The third objective is to build an operational prompt to produce an LPP/GORGIAS annotation in brat format (.ann), that is, an LPP structured representation directly in a .ann file. The fourth objective is to evaluate this prompt on two complementary texts: the case *Alka v. Turkey*, for legal reasoning, and *Twelve Angry Men*, for deliberative reasoning. The evaluation covers the .ann files, the .png graphs, and the observed errors. Finally, the fifth objective is to identify the limits of automation and the points that require human validation.

1.4 Report Organization

Chapter 2 presents formal argumentation, computational frameworks, and the LPP/GORGIAS formalism. Chapter 3 constitutes the main contribution: corpus analysis, annotation principles, prompt construction and evaluation, followed by a discussion of results. Chapter 4 synthesizes the contributions of the TER, addresses the research question, and opens perspectives for the continuation of the GRAIL project.

Chapter 2

Related Work

2.1 Formal Argumentation and Computational Frameworks

Computational argumentation models reasoning in which arguments, built from premises and conclusions, can support or attack one another. The goal is to determine which arguments remain acceptable when they come into conflict.

Dung’s framework [2] constitutes a major foundation of the field. It represents an argumentative system as a pair $(Args, Att)$, where $Args$ is a set of arguments and Att an attack relation. Its acceptability semantics, notably admissible, preferred, and stable, allow the selection of coherent sets of arguments. This framework is general and robust, but remains abstract: it describes neither the premises, nor the rules, nor the conclusions internal to the arguments.

ASPIC+ [3] partially addresses this limitation by structuring arguments. It introduces strict and defeasible rules, distinguishes several forms of attacks, and allows the use of preferences between rules. It thus brings argumentative reasoning closer to a more exploitable logical representation, while remaining distinct from a textual annotation scheme.

Argument mining [4] deals precisely with this extraction from natural texts. Its main tasks are:

- identifying argumentative components, such as **Claim** and **Premise**;
- recognizing relations such as **Support** and **Attack**;
- building or exploiting annotated corpora to train and evaluate automatic systems.

These corpora remain heterogeneous. Legal texts, argumentative essays, and online discussions do not use the same units, labels, or granularity. Argument mining resources are therefore useful for identifying argumentative structures, but they are not sufficient to produce decision-making reasoning with explicit priorities. This is the contribution of LPP/GORGIAS: distinguishing rules, preferences between rules, and preferences between preferences.

2.2 Logic Programming with Priorities (LPP) and GORGIAS

LPP, for *Logic Programming with Priorities*, organizes an agent's reasoning into three levels. Its formal structure is defined in [5]:

Definition 4 [5]. *An agent's argumentative policy theory or theory, $\mathcal{T}$, is a triple $\mathcal{T} = (T, P_R, P_C)$ where the rules in T do not refer to h_p , all the rules in P_R are priority rules with head $h_p(r_1, r_2)$ s.t. $r_1, r_2 \in T$ and all rules in P_C are priority rules with head $h_p(R_1, R_2)$ s.t. $R_1, R_2 \in P_R \cup P_C$.*

Thus T contains object-level rules deriving Options from Context, P_R priority rules expressing which rule takes precedence (via h_p), and P_C meta-priority rules expressing which priority itself takes precedence. Conflicts are not resolved by priority rules alone: the admissibility semantics exploits h_p to select acceptable arguments.

Based on this definition, this TER uses the labels **rule** (for T), **prefer** (for P_R) and **meta_prefer** (for P_C), as summarised below.

Definition 4 [5]	TER label	Head of the rule
T object-level rules	rule	any literal (conclusion / Option)
P_R priority rules	prefer	$h_p(r_1, r_2)$, with $r_1, r_2 \in T$
P_C meta-priority rules	meta_prefer	$h_p(R_1, R_2)$, with $R_1, R_2 \in P_R \cup P_C$

Table 2.1: Correspondence between Definition 4 [5] and the annotation labels used in this TER

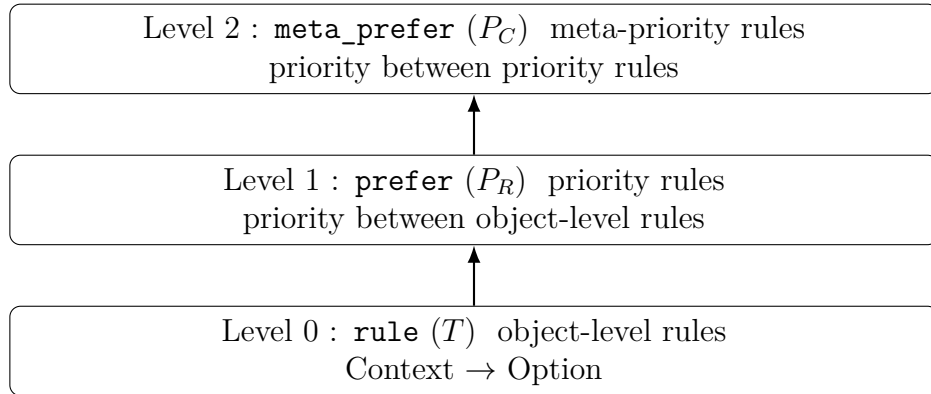


Figure 2.1: Structure of the three LPP levels

GORGIAS [1] is the Prolog implementation of this framework. Each rule has the form **name** = {cond} $\vdash$ **head**, where $\vdash$ denotes defeasible derivation and the level in $\mathcal{T} = (T, P_R, P_C)$ is determined by **head**: a plain conclusion for T (**rule**); $r_1 > r_2$ with $r_1, r_2 \in T$ for P_R (**prefer**); $R_1 > R_2$ with $R_1, R_2 \in P_R \cup P_C$ for P_C (**meta_prefer**). The names **a**, **pc**, **cw**, **d**... are modeler conventions: in the shopping assistant example (see listing and Fig. 2.2 below), **a**(X) belongs to T , **pc**(X) to P_R , and **cw/cs/cl/d** to P_C .

```

1  OPTIONS = {buy(lamb), buy(pork), buy(chicken), buy(fish)}
2
3  a(pork) = {main_shopping} |- buy(pork)
4  a(lamb) = {main_shopping} |- buy(lamb)
5  a(chicken) = {main_shopping} |- buy(chicken)
6  a(fish) = {main_shopping} |- buy(fish)
7
8  pc1(pork) = {cheap(pork)} |- a(pork) > a(lamb)
9  pc2(pork) = {cheap(pork)} |- a(pork) > a(fish)
10 pc1(chicken) = {cheap(chicken)} |- a(chicken) > a(lamb)
11 pc2(chicken) = {cheap(chicken)} |- a(chicken) > a(fish)
12 pc3(pork) = {cheap(pork)} |- a(pork) > a(chicken)
13 pc3(chicken) = {cheap(chicken)} |- a(chicken) > a(pork)
14
15 cw(pork) = {winter} |- pc3(pork) > pc3(chicken)
16 cs(chicken) = {summer} |- pc3(chicken) > pc3(pork)
17 cl(chicken) = {local(chicken)} |- pc3(chicken) > pc3(pork)
18
19 d(chicken) = {} |- cl(chicken) > cw(pork)

```

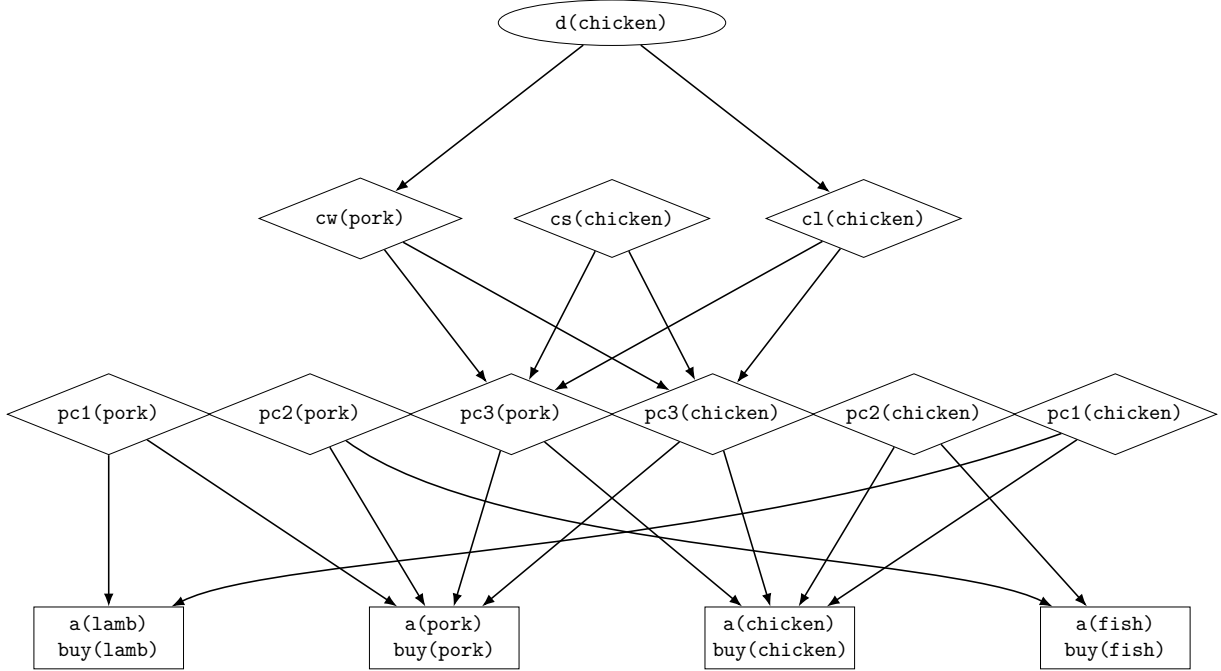


Figure 2.2: Complete GORGAS shopping assistant example

GORGAS has been applied to several domains: medical support, ophthalmic clinic assistant, network security, firewall configuration, business computing, ambient intelligence, cognitive personal assistants, medical data access and sharing, and investment tools such as PORTRAIT. The paper also presents Gorgias-B, which generates executable GORGAS code from scenario-based preferences. These applications confirm the relevance of LPP for decisions based on incomplete information and conflicting priorities, which aligns with the objective of the GRAIL project.

Chapter 3

Contribution

3.1 Corpus Resource Analysis

3.1.1 General Overview of the Corpus

The corpus provided by the supervisors of this research project brings together 17 research datasets on argumentative annotation, accompanied by papers and guidelines. It covers biomedical, legal, deliberative, scientific, and political texts, argumentative essays, and online discussions. It constitutes a panorama of argument mining useful to the GRAIL project, but it is not directly annotated in the LPP format.

Most resources describe **Claim**, **Premise**, **Support** and **Attack**, without making explicit the levels **rule**, **prefer** and **meta_prefer**. The challenge is therefore not to mechanically convert existing labels, but to evaluate what each dataset allows to reconstruct for LPP/GORGIAS (see the GitHub repository (Appendix)).

Theme	LPP Relevance	LPP Limitation
Legal	Decisions, facts, applicability of rules	Priorities rarely explicit
Deliberative	Conflicts, opposing positions, changes of opinion	High degree of human interpretation
Argumentative essays	Claim , Premise , Support easily identifiable	Decisional reasoning rarely explicit
Online discussions	Numerous interactions and oppositions	Fragmented structure, prefer absent
Benchmarks and mixed formats	Model comparison and standardized formats	Reinterpretation required for LPP/GORGIAS

Table 3.1: Overview of the dataset families studied

The observed formats are heterogeneous: .txt and .ann for ArgumentMiningECHR and NoDE_datasets, XML for the arg-microtexts, CSV or TSV for Dataset-master and ARIES-Benchmark, and papers and guidelines to clarify the semantics of the labels.

3.1.2 Analysis Criteria for LPP

Reading the GORGIAS paper led us to evaluate the datasets according to their capacity to represent decisional reasoning, and not merely general argumentation. A text may contain **Claim** and **Premise** without allowing the reconstruction of Context, Option, rule, prefer and meta_prefer.

Five criteria guided the analysis:

- **Conclusions or decisions:** presence of verdicts, judgments, recommendations or positions that can become Option.
- **Conditions or facts:** presence of circumstances, observations or established facts that can become Context.
- **Condition-conclusion relations:** possibility of reconstructing rule, that is, a logical link between conditions and conclusion.
- **Conflicts between conclusions:** existence of opposing arguments or rules in tension, a necessary condition for prefer.
- **Explicit priorities:** presence of textual markers indicating that one rule or conclusion prevails over another, the most discriminating condition for prefer and meta_prefer.

3.1.3 Exploitable Information for LPP Annotation

Legal datasets are the most directly suited for the transition to LPP. ArgumentMining-ECHR [6] contains clear decisional reasoning: established facts determine the applicability of an article, then lead to a conclusion of violation or non-violation. The .ann files provide a basis for identifying relevant segments. The main limitation concerns priorities: legal reasoning often produces chains of rule, but rarely makes the hierarchy between two competing rules explicit.

Deliberative resources, in particular Twelve Angry Men in NoDE_datasets [7], are rich but more interpretive. They contain:

- asserted and contradictory positions;
- explicit conflicts between speakers;
- potential priority markers;
- changes of position and reformulations of alternatives.

They therefore allow us to test the robustness of the LPP schema, but require more human validation, as no court or institution formulates a single final conclusion.

Argumentative essays are less suitable. They often contain **Claim**, **Premise** and **Support** structures, but without explicit decisional reasoning or priorities between rules. They can help identify potential Context and Option, but remain insufficient for constructing prefer or meta_prefer.

Online discussions are the least directly exploitable. Reasoning is fragmented, positions are sometimes poorly formalized, and **Support** or **Attack** relations do not automatically convert into rule or prefer.

3.1.4 Limitations of Existing Argumentative Annotations

The first limitation is the heterogeneity of labels, granularity and guidelines. No standard is directly compatible with LPP, and the same label may change in scope depending on the corpus or domain.

Correspondences with LPP must therefore be carefully controlled:

- a **Claim** is not necessarily an Option: it may be an intermediate premise or a local conclusion;
- a **Premise** is not necessarily a Context: it may be the conclusion of a preceding argument;
- a **Support** relation is not necessarily a **rule**, as it does not always specify the Condition and the Effect;
- an **Attack** relation is not a complete LPP conflict: it is not sufficient to create a **prefer** without an explicit priority.

Three structural limitations explain the need for a specific schema:

- **implicit preferences**: priorities are often implied, whereas LPP requires structured preferences;
- **absent relations**: many annotations identify the components without reconstructing the chains of reasoning;
- **unstable granularity**: some spans are too broad to become Context or Option, others too fragmentary to retain argumentative value.

Existing annotations are therefore a basis for identification, but not an LPP representation. The corpus analysis shows that **Claim**, **Premise**, **Support** and **Attack** do not directly determine Context, Option, **rule**, **prefer** and **meta_prefer**. The transition to GORGIAS requires dedicated annotation principles to move from natural text to a usable LPP structure, and then to construct the annotation prompt.

3.2 Annotation Principles for the Transition to LPP

3.2.1 Retained Schema

The retained schema corresponds to the strict core of LPP, translated into annotatable elements in the brat format [8]. It draws on the .ann formats observed in the corpus, without directly reusing the classical labels of argument mining, in order to link a natural text to a representation compatible with LPP/GORGIAS.

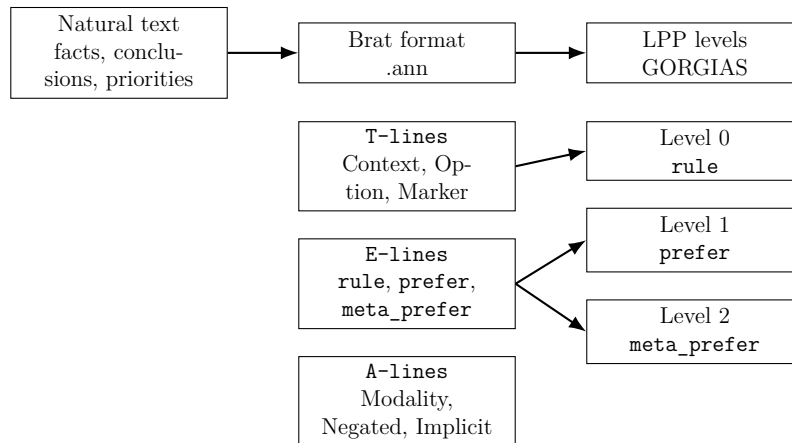


Figure 3.1: Correspondence between natural text, brat annotations and LPP levels

The schema comprises three textual entities, annotated by **T-lines**, and three events, annotated by **E-lines**. The entities identify the relevant segments of the text; the events structure these segments according to LPP levels.

The schema also comprises three attributes annotated by **A-lines**: **Modality** applies exclusively to Options and remains optional with the values **obligatory**, **permitted**, **recommended**, or **default**; **Negated=True** is reserved for semantic negations on which the reasoning depends; **Implicit=True** flags an invented span used to complete an otherwise incomplete reasoning and must remain exceptional.

3.2.2 Annotation Principles

Annotation follows a stable working order:

- Start with conclusions: verdicts and decisions in formal texts, positions asserted or contested in deliberative texts.
- Verify that an Option is endorsed by the speaker: a rhetorical question or a reported argument is not sufficient.
- Then move back to the Context elements that support the Options, using short but semantically complete spans.
- Reuse the same T identifier when the same fact or the same option recurs multiple times.
- Group into a single **rule** all Conditions that jointly support the same conclusion.
- Chain **rule** events when an intermediate conclusion becomes the Condition of a final conclusion, a pattern common in legal texts.

The creation of preferences is more restrictive than that of **rule** events. A **prefer** presupposes two already identified conflicting **rule** events and an explicit priority Marker; a simple “however” or “but” is not sufficient.

A **meta_prefer** is limited to three cases:

- a more specific context overturns the result of an existing **prefer**;
- a speaker explicitly abandons or reverses a prior preference;
- a speaker explicitly declares that, under a future condition, a different preference applies.

The following table summarises the role of each element and the restrictions applied during annotation.

Element	Role in annotation	Main restrictions
Context	Fact, condition, circumstance, exception or procedural posture feeding a conclusion.	Exclude reformulations of an Option, already annotated consequences, and rhetorical asides.
Option	Conclusion, decision, judgement, asserted position or available alternative.	Exclude abstract concepts, rhetorical questions, and reported speech not endorsed by the speaker. If a sentence expresses a complete preference, annotate the alternatives separately.
Marker	Explicit expression of priority: “prevails over”, “takes precedence over”, “is preferred to”, “notwithstanding”, “subject to”.	“however”, “but”, “yet”, or “nevertheless” are not sufficient.
rule	Level 0: one or more Conditions for exactly one Effect.	Effect = Option; conjoint Conditions of the same conclusion within a single rule .
prefer	Level 1: conflict between two rule events.	The two conflicting rule events must be identified beforehand. Winner/Loser point to E (events), not T. Textually explicit priority is mandatory.
meta_prefer	Level 2: conflict between preferences.	Winner/Loser = prefer or meta_prefer , never rule (no level-skipping).
Negated=True	Semantic negation whose positive counterpart plays a distinct argumentative role.	Reserve for central semantic negations; exclude procedural negations.

Table 3.2: Elements of the brat annotation schema for LPP/GORGIAS

On the technical side, spans must delimit precise segments: avoid overly broad spans (mixing several pieces of information) and truncated spans (meaning lost out of context). Brat offsets follow the convention of zero-based start, exclusive end.

3.3 Construction and Evaluation of an AI Tool for LPP Annotation

3.3.1 Prompt Design

The prompt constitutes the operational version of the annotation principles described previously. It is written in English to maximize the performance of the language model used, while remaining applicable to any argumentative text.

The prompt is organized around eight blocks:

- LPP reminder: three levels, linked to `rule`, `prefer` and `meta_prefer`;
- annotation schema: Context, Option, Marker, events and roles Condition, Effect, Winner, Loser, When;
- preference texts: explicitly hierarchized alternatives, without equating rhetorical opposition with priority;
- methodology: conclusions, conditions, `rule`, then `prefer` and `meta_prefer` if conditions are met;
- errors to avoid: overly wide spans, rhetorical questions as Option, preferences based on simple connectors;
- examples: a three-level shopping assistant, a legal text on the presumption of innocence, a dialogue close to Twelve Angry Men, a case with `prefer` and `meta_prefer`;
- output format: T-lines, `rule`, `prefer`, `meta_prefer`, A-lines, with sequential identifiers T1, E1, A1;
- self-verification: truncated spans, duplicates, LPP levels, offsets; the response must start with T1, without any introduction.

Each `rule` must begin with `rule:T...`, contain at least one Condition and exactly one Effect. The prompt explicitly requests not to produce `prefer` or `meta_prefer` when the text does not contain an explicit priority. The complete prompt is provided in the GitHub repository (Appendix).

3.3.2 Output Format: the .ann File

The standard brat format [8] was retained because it is compatible with several existing resources, notably ArgumentMiningECHR [6], readable by a human annotator and transformable into a .png graph.

An .ann file produced by the prompt contains T-lines for entities, E-lines for events and A-lines for attributes:

```
1 T1 Context 729 816 the applicant's grievances...
2 E1 rule:T4 Condition:T2 Condition:T3 Effect:T4
3 E21 prefer:T71 Winner:E2 Loser:E9 When:T69
4 A1 Modality T5 obligatory
```

The .ann file can then be transformed into a visualization graph in .png format by a dedicated Python script; it directly constitutes the LPP/GORGIAS representation in brat format, without conversion to an intermediate .lpp file.

3.3.3 Qualitative Evaluation of the Prompt

The prompt was evaluated on two complementary texts: the *Alka v. Turkey* judgment, which involves formal legal reasoning, and a deliberative excerpt from *Twelve Angry Men* [7], which involves multiple speakers. The complete .ann files are reproduced in the GitHub repository (Appendix), and the complete .png graphs are provided in the GitHub repository (Appendix).

For the *Alka v. Turkey* judgment, the .ann file obtained contains 28 entities, 14 **rule** and 7 modality attributes, with several multi-step chains including $E1 \rightarrow E2 \rightarrow E3 \rightarrow E4$, from the existence of a link between proceedings to the admissibility of the complaint.

No **prefer** or **meta_prefer** is produced, since the text does not explicitly contain any priority between rules. The modalities notably concern the applicability of Article 6 § 2, admissibility, the possibility for the civil court to reach a different conclusion, and the final finding of a violation. The LPP graph is linear and illustrates an essentially deductive reasoning.

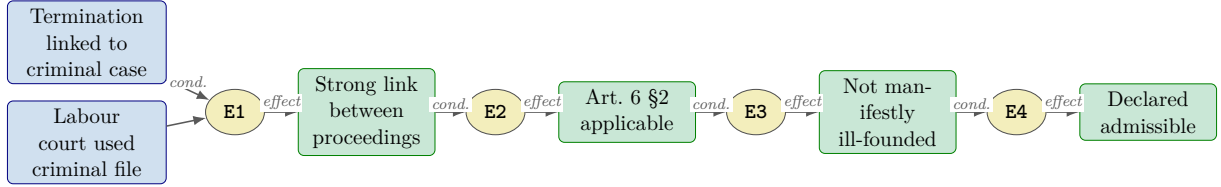


Figure 3.2: Chain $E1 \rightarrow E2 \rightarrow E3 \rightarrow E4$ in the *Alka v. Turkey* annotation: multi-step deductive sequence leading to the admissibility of the application.

This structure is consistent with the legal text, where the argumentation progresses through applicability, admissibility and then conclusion, and confirms that the prompt did not introduce any preference hierarchy absent from the text.

For *Twelve Angry Men*, the .ann file is denser: 74 entities, 20 **rule**, 1 **prefer** and 12 attributes including 4 **Negated=True**.

The single **prefer**, E21, opposes two concrete **rule** events. E2, Winner, argues that the boy is not guilty until the jury has decided so; E9, Loser, argues that one cannot believe what the targeted group says. The Context T69 and the Marker T71 both correspond to the expression “you can’t trust them”. This result shows that the prompt does not automatically convert every disagreement into an LPP preference, but it must be reviewed: T71 is more of a generalization about credibility than a canonical priority marker.

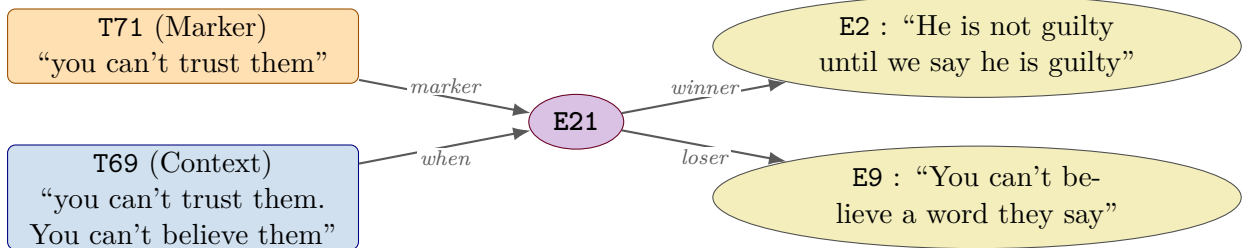


Figure 3.3: Structure of **prefer** E21 in the *Twelve Angry Men* annotation: E2 (presumption of innocence) takes precedence over E9 (credibility attack) via the marker “you can’t trust them” (T69, T71).

The fact that T69 and T71 both reproduce “you can’t trust them” reveals an ambiguity: the text expresses distrust, but not necessarily a normative priority, which illustrates both the prompt’s ability to isolate a conflict and the necessity of human validation.

The following comparative table summarizes the differences observed between the legal text and the deliberative text.

Criterion	Alka v. Turkey	Twelve Angry Men
Text type	Legal	Deliberative
Annotated entities	28: 13 Context, 14 Option, 1 Marker	74: 40 Context, 33 Option, 1 Marker
rule	14 rules	20 rules
prefer	None	1 event, E21
meta_prefer	None	None
Attributes	7 Modality attributes	12 attributes: 8 Modality, 4 Negated
Graph structure	Linear, deductive	Parallel branches, multi-speaker
Main errors	Spans T14 and T16 too wide; empty Marker T25	Rhetorical question T15 annotated as Option; proximity between T27 and T28; debatable Marker T71

Table 3.3: Qualitative comparison of the two annotated texts

The quality of the annotation therefore depends strongly on the nature of the text: explicit decisional reasoning, as in Alka, produces cleaner structures than a deliberative text, which is richer but more interpretive.

3.3.4 Limitations of Automation and Human Validation

The prompt produces usable annotations on both types of texts, but human review remains necessary to validate interpretive choices: genuinely held versus rhetorical position, textually grounded **prefer**, semantically complete spans.

Three factors determine annotation quality:

- text clarity, since an explicit decisional structure yields better results;
- the presence of explicit markers, without which no **prefer** can be correctly created;
- the prompt’s adherence to LPP schema constraints, which avoids creating textually ungrounded **prefer** or **meta_prefer**; any deviation introduces a reasoning hierarchy absent from the text.

The prompt must remain general and applicable to any type of argumentative text; specific corrections must be applied manually to the .ann file.

Chapter 4

Conclusion

4.1 Summary of Contributions

This TER proposed a method to guide the transition from argumentative text to an LPP representation compatible with GORGIAS, based on a corpus of 17 argument mining datasets provided as part of the ANR GRAIL project. Three main contributions were made:

- **Corpus analysis:** evaluation of the 17 datasets according to their relevance for LPP; identification of usable information and limitations of existing schemas; selection of ArgumentMiningECHR and Twelve Angry Men as the most suitable resources, for their respectively formal legal and multi-speaker deliberative reasoning.
- **Annotation principles:** definition of a brat schema centred on Context, Option, Marker, **rule**, **prefer** and **meta_prefer**, accompanied by precise criteria to avoid false positives (**Claim** $\neq$ Option, **Support** $\neq$ **rule**, rhetorical opposition $\neq$ **prefer**) and constraints on span granularity and modality attributes.
- **Prompt and evaluation:** construction of an operational prompt in English producing brat annotations at three LPP levels, evaluated on two complementary texts (an ECHR ruling: 14 **rule**, no **prefer**; a deliberative excerpt: 20 **rule**, 1 **prefer**), with generation of .ann files and .png visualisation graphs.

These contributions show that existing argumentative annotations (**Claim**, **Premise**, **Support**, **Attack**) constitute a useful identification base, but not a directly usable LPP representation. The transition to GORGIAS requires dedicated structuring, assisted here by a prompt.

4.2 Answer to the Research Question

The central research question was as follows: how can we guide the transition from argumentative text to an LPP representation compatible with GORGIAS, and to what extent can this transition be automated? The results show that such a transition is possible provided the expected levels are explicitly defined. Annotation must go beyond the extraction of general argumentative components to reconstruct a logical structure where Contexts support Options through **rules**, then, if the text warrants it, through **prefer** or **meta_prefer**.

Level 0 is the most accessible: *Alka v. Turkey* illustrates this with well-formed chains of **rule**, without artificial priority. Level 1 is more delicate, as it presupposes two conflicting **rules** and a textual cue of priority; in *Twelve Angry Men*, the produced **prefer** remains to be validated, as its Marker pertains more to a generalisation about credibility than to an explicit priority marker. It should be noted that requiring an explicit textual Marker is a methodological annotation choice adopted in this TER, not a formal constraint of the LPP formalism, which does not impose textual anchoring for preferences. Level 2 is the rarest: no **meta_prefer** was produced in either of the two evaluated texts.

Automation helps to homogenise annotation and produce a first usable version, but it does not alone guarantee the validity of an LPP representation. Human validation remains necessary to control the boundary between Context and Option, verify **prefer** and **meta_prefer**, correct spans and offsets, and assess the legitimacy of interpretations. Existing corpora are therefore useful but insufficient: they must be reinterpreted to produce LPP structures executable by GORGIAS.

4.3 Perspectives

The prompt should be tested on medical, political, and institutional texts in order to distinguish cases where LPP is appropriate from those where the text is too rhetorical or implicit. A manually annotated reference set would allow recurring errors to be measured and several versions of the prompt to be compared. On the technical side, it remains to automatically validate .ann files and to convert annotations into .lpp files executable in GORGIAS. In the longer term, the schema and prompt developed in this TER are intended to serve as a foundation for annotated corpora in LPP format within the GRAIL project, from natural text to a GORGIAS representation usable by an argumentative agent.

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Statement on the Use of Artificial Intelligence

Artificial intelligence tools were used in this project only within the scope explicitly authorized by our supervisors and in accordance with the guidelines they provided for their use.

The platform PLEIADE MATH-INFO (<https://pleiade.mi.parisdescartes.fr>) was used exclusively to translate short excerpts taken from the GRAIL folder. These excerpts are not public, and the use of this platform was strictly limited to translation support for those materials.

Claude and ChatGPT were also used to support the reading of certain research papers that are already publicly available. In this context, these tools were used to clarify concepts, reformulate technical passages, and facilitate understanding, but not to replace the reading, interpretation, or critical analysis of the original publications.

For the AI-Based Approach presented in this report, which was explicitly requested by our supervisors, several AI systems were used in a controlled experimental setting. Claude Code was used to help create scripts intended to transform annotation files into graph structures. Claude Opus, Claude Sonnet, and ChatGPT 5.5 Thinking were used both for prompt design and for the execution of the prompts developed as part of this approach.

In order to reduce contextual bias and avoid influencing the outputs, all prompt tests were conducted in fresh conversations with no prior interaction history.

All copyright in this report, as well as the entirety of its intellectual property, remain with the authors.

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Appendix

Code and Annotation Resources

The scripts, annotation files, and resources produced during this TER are available in the following GitHub repository:

<https://github.com/thmsgo18/lpp-argument-mining>

The repository contains:

- the complete annotation prompt used to produce the LPP/GORGIAS annotations;
- the Python script used to convert `.ann` annotation files into `.png` visualisation graphs;
- the raw texts and full results (`.ann` files and `.png` graphs) for the two analysed texts: the Alka v. Turkey ruling and the Twelve Angry Men excerpt.